

# Midterm (L3 - Ms. Maria Samad)

CS 101 Algorithmic Problem Solving

Fall 2023

Name(s): \_\_\_\_\_

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“All images collected from Google”

## 1. Long Weekend

The best time to visit Laurentian Mountains is in Autumn, as you can see all the beautiful colors of the season over there.



Figure 1: Fall in Laurentian Mountains

Mr. AA decides to take his family for a visit to the mountains, and books a stay at the MT Resort. The resort offers breakfast, lunch and dinner buffets for extra charges, based on the age of the guest. However, not everyone in the family will avail all the buffets in a day. For example, one person may just go for breakfast and dinner, and another may go for all three, etc.

It is becoming difficult to plan the budget of the trip. Hence, Mr. AA decides to note down all the possible buffet options his family members are interested in, and plan a daily budget for each person accordingly.

Help Mr. AA plan his daily food budget per person, according to the food choices specified by each member of his family.

The buffet charges are as follows:

|                            | Breakfast | Lunch | Dinner |
|----------------------------|-----------|-------|--------|
| Adult (ages 13 and above)  | \$20      | \$30  | \$35   |
| Child (ages 6 to 12)       | \$10      | \$20  | \$20   |
| Toddler (ages 5 and below) | FREE      | FREE  | \$5    |

Given the “*age*” of the family member, and their choices of breakfast, lunch and dinner as “*b*”, “*l*”, and “*d*”, respectively, it calculates the daily food charges of that individual and prints the output, as mentioned in the Sample.

### Constraints

- $age \in \mathbb{N}$
- $age > 0$
- $b \in \{'Y', 'y', 'N', 'n'\}$
- $l \in \{'Y', 'y', 'N', 'n'\}$
- $d \in \{'Y', 'y', 'N', 'n'\}$

### Interaction

The input comprises of two lines. The first line contains the *age* of the family member. The second line contains three single-spaced letters, *b*, *l*, and *d*  $\in \{'Y', 'y', 'N', 'n'\}$  representing Buffet, Lunch and Dinner choices, respectively (‘Y’ or ‘y’ if they choose to select the option, and ‘N’ or ‘n’ if they don’t select it).

The output contains a string statement giving the total charges for each member. Note that for an adult the statement uses “Adult”, for a child the statement prints “Child” and for toddler, it prints “Toddler”, depending on the age of the guest.

### Sample

| Input             | Output                                  |
|-------------------|-----------------------------------------|
| 28<br>'Y' 'y' 'N' | Adult charges for the day will be: \$50 |
| 10<br>'y' 'n' 'N' | Child charges for the day will be: \$10 |

In the first case, the age of the guest is 28, which means the guest is an adult. The guest has selected breakfast and lunch buffets only, but no dinner buffet. The total charges will then be:  $\$20 + \$30 = \$50$ . Hence the statement is printed with “Adult” in the text along with the charges.

In the second case, the age of the guest is 10, which means the guest is a child. The guest has selected only the breakfast buffet. The total charges will then be: \$10. Hence the statement is printed with “Child” in the text along with the charges.

**Exercise**

In the space provided, indicate the outputs for the given inputs.

| Input       | Output |
|-------------|--------|
| 35          |        |
| 'Y' 'N' 'Y' |        |
| 5           |        |
| 'y' 'y' 'y' |        |
| 12          |        |
| 'n' 'N' 'Y' |        |

**Problem Identification**

Briefly explain the underlying problem you identified in the above question that led you to your solution.

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**Pseudocode**

**Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

**2. Chocolate Distribution**

Professor Mars assigned some tasks with a deadline to each student. However, students have not been completing their tasks enthusiastically.



Figure 2: Chocolates

In order to motivate the students more, Professor Mars decide to give chocolate bars to students based on their average daily performance as follows:

- Less than 40% - no chocolate awarded
- [40% – 50%) - 1 chocolate bar
- [50% – 65%) - 2 chocolate bars
- [65% – 80%) - 3 chocolate bars
- [80% – 100%) - 4 chocolate bars
- 100% and above - 5 chocolate bars

Remember, square brackets in intervals mean inclusive, and round brackets mean not included.

Given “ $N$ ” number of students in class, each of whom has completed “ $n$ ” tasks in “ $m$ ” number of days, it calculates the number of chocolates, and prints the output as shown in Sample.

### Constraints

- $1 \leq N \leq 25$
- $1 \leq n \leq 10$
- $1 \leq m \leq 15$
- Use loops to solve the problem.

### Interaction

The input comprises of multiple lines. The first line contains the number of students in class,  $N$ . The following  $N$  lines contain two space-separated integers representing  $n$  - the number of tasks assigned, and  $m$  - the number of days to complete the task, for each student.

The output contains a string statement giving the total number of chocolates that Professor Mars need to buy.

### Sample

| Input                                                                          | Output                                                               |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 4<br>2 3<br>5 5<br>6 14<br>3 6                                                 | Professor Mars will need to buy 11 chocolates to distribute in class |
| 10<br>1 3<br>6 1<br>10 10<br>3 7<br>5 12<br>4 6<br>9 5<br>10 15<br>2 9<br>7 13 | Professor Mars will need to buy 25 chocolates to distribute in class |

In the first case, there are 4 students in class. First one did 2 tasks in 3 days, so their performance is 66.67%, which means this student gets 3 chocolates. Second one completed 5 tasks in 5 days, so their performance is 100%, which means this student gets 5 chocolates. Third one completed 6 tasks in 14 days, so their performance is 42.87%, which means this student gets 1 chocolate. Last one did 3 tasks in 6 days, so their performance is 50%, which means this student gets 2 chocolates. The total number of chocolates will be  $3 + 5 + 1 + 2 = 11$  chocolates.

In the second case, there are 10 students. First student gets no chocolate, second and third students get 5 chocolates each, fourth and fifth students get 1 chocolate each, sixth gets 3 chocolates, seventh one gets 5, eight gets 3, ninth gets none, and last gets 1. The total number of chocolates will be  $0 + 5 + 5 + 1 + 1 + 3 + 5 + 3 + 0 + 1 = 25$  chocolates.

**Exercise**

In the space provided, indicate the outputs for the given inputs.

| Input | Output |
|-------|--------|
| 5     |        |
| 3 2   |        |
| 6 4   |        |
| 9 6   |        |
| 12 8  |        |
| 1 10  |        |
| 7     |        |
| 2 15  |        |
| 3 6   |        |
| 5 12  |        |
| 6 3   |        |
| 8 9   |        |
| 9 1   |        |
| 1 1   |        |
| 3     |        |
| 5 5   |        |
| 6 6   |        |
| 7 7   |        |

**Problem Identification**

Briefly explain the underlying problem you identified in the above question that led you to your solution.

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**Pseudocode**



**Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

**3. Matherns**

Sarah loves Mathematics, and plays all the mobile/computer games based on numbers. A latest game called “Matherns” has been introduced in the market, in which the players are supposed to code the logic for the given pattern of numbers.

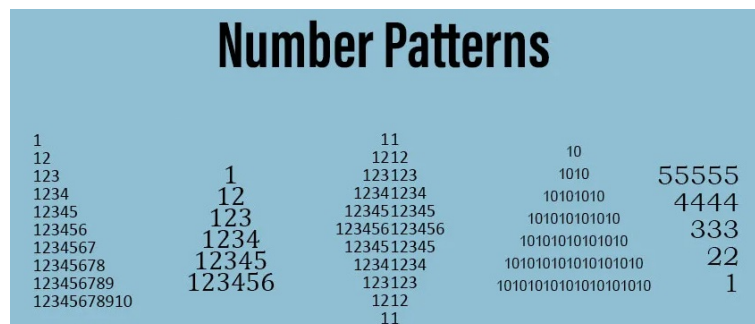


Figure 3: Patterns are FUN!

The player is given a single number which represents the number of rows that must be in the pattern, and using that number, they should code the logic to achieve the given pattern (See Sample Outputs)

Sarah decided to ask for help from her elder brother who is a Computer Scientist to understand the logic, and he provides his feedback on whether she is correct or not.

Given  $n$  input, it prints a pattern for line numbers starting at  $n$  and going all the way to 1.

**Constraints**

- $n \in \mathbb{N}$
- $n > 0$

**Interaction**

The input contains a single integer,  $n$  and using that  $n$  value, the patterns can be printed.  
The output is the desired number pattern printed on screen.

**Sample**

| Input | Output                                                                             |
|-------|------------------------------------------------------------------------------------|
| 5     | 5 25 125 625 3125<br>4 16 64 256<br>3 9 27<br>2 4<br>1                             |
| 6     | 6 36 216 1296 7776 46656<br>5 25 125 625 3125<br>4 16 64 256<br>3 9 27<br>2 4<br>1 |

In the first case, the input is 5, so it prints 5 lines of patterns, with each line's pattern containing numbers that are the successive powers of line number.

In the second case, the input is 6, so it prints 6 lines of patterns, with each line's pattern containing numbers that are the product of the previous number and the line number.

**Exercise**

In the space provided, indicate the outputs for the given inputs.

| Input | Output |
|-------|--------|
| 4     |        |
| 7     |        |
| 2     |        |

**Problem Identification**

Briefly explain the underlying problem you identified in the above question that led you to your solution.

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**Pseudocode****Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

**4. Golden Wedding**

Mr. and Mrs. Weasley have been happily married for 50 years, and their children decided to celebrate their golden wedding anniversary on a grand scale.



The Weasleys



Hogwarts Lake

Figure 4: Golden Wedding Anniversary of Mr. & Mrs. Weasley

The Weasley children have planned a weekend getaway for their parents and the guests at Hogwarts Lake. As it is a very popular tourist spot, the reservation to the Hogslake Resort must be made well in advance. The guests are expected to arrive on Friday night and leave on Sunday, i.e. it will be a 2-night stay.

In order to book the rooms for the guests, they searched online to get the following information about the available rooms at the resort:

| Room Description                    | Room Code | Cost Per Night |
|-------------------------------------|-----------|----------------|
| Single room with one single bed     | "1PR"     | £115.11        |
| Small family room for 2 to 3 guests | "SFR"     | £235.23        |
| Large family room for 4 to 5 guests | "LFR"     | £340.34        |

Help the Weasley kids to calculate the total cost of the booked rooms by writing a function called:

- *GetRoomCost* that takes two arguments: *num\_rooms*, which is an integer representing the number of rooms, and *room\_code*, which is a string, as mentioned in the table above.
- This function uses a helper function called *GetRoomRate* that takes the *room\_code* as argument, and returns its cost per night.

After getting the cost per night, calculate the total cost of that type of room, and return the value.

Call this function three times for the three inputs representing the three types of rooms, i.e. "*s*" for number of single rooms, "*sf*" for number of single family rooms, and "*lf*" representing the number of large family rooms, and then get the total number of rooms and the total cost of all the rooms to be booked.

### Constraints

- $s, sf, lf \in \mathbb{N}$
- $s, sf, lf > 0$

### Interaction

The input contains a three single-spaced integers, *s*, *sf*, and *lf* representing the number of single rooms, small family rooms and large family rooms respectively.

The output prints the total number of rooms and their total cost in the format specified in the Sample Output.

**Sample**

| Input | Output                                                   |
|-------|----------------------------------------------------------|
| 5 2 2 | Rooms required: 9<br>All the rooms will cost: £3453.38   |
| 9 9 9 | Rooms required: 27<br>All the rooms will cost: £12432.24 |

In the first case, the input is 5 single rooms, 2 small family rooms, and 2 large family rooms. Each single room costs £115.11 per night, so 5 rooms will cost £575.55 for one night. Each small family room is £235.23 per night, so 2 rooms will cost £470.46 for one night. One large family room costs £340.34 per night, so for 2 rooms will cost £680.68 per night. The total for one night will be  $£575.55 + £470.46 + £680.68 = £1726.69$ , however, the guests will be staying for two nights, so the total cost will be £3453.38; and the total number of rooms will be  $5 + 2 + 2 = 9$ , which is what is printed in the output

In the second case, the inputs are 9 single rooms, 9 small family rooms, and 9 large family rooms. Using the same logic as explained in the first case, the total number of rooms will be  $9 + 9 + 9 = 27$ , and the total cost of these 27 rooms for 2 nights will be £12432.24, and output is printed as desired

**Exercise**

In the space provided, indicate the outputs for the given inputs.

| Input | Output |
|-------|--------|
| 1 2 3 |        |
| 7 9 5 |        |
| 2 6 8 |        |

**Problem Identification**

Briefly explain the underlying problem you identified in the above question that led you to your solution.

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**Pseudocode****Dry Run**

Using any one of the inputs provided in the **Exercise** section above, dry run your pseudocode in the space below.

## TIME TO DO SOME DEBUGGING!

### 5. Percy Jackson & the Battle of Labyrinth

According to Greek mythology, the Labyrinth was an elaborate, confusing structure designed and built by the legendary artificer Daedalus for King Minos. Its function was to hold the Minotaur, the monster eventually killed by the hero Theseus. Daedalus had so cunningly made the Labyrinth that he could barely escape it after he built it. Percy Jackson and his friends are going on a quest to find Daedalus in the Labyrinth. In order to successfully reach the center of Labyrinth and coming out, they need to overcome several obstacles.

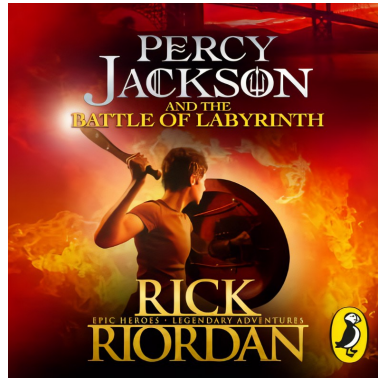


Figure 5: Percy Jackson & the Battle of Labyrinth

Their final task requires them to have some basic knowledge of Mathematics and programming logic. The only thing that is stopping them now from coming out is a huge rock, that can only be moved by using the numbers on the rock.

Help them solve the task, so they can complete their quest successfully, by writing a function called *solving\_labyrinth(n)*. The function takes  $n$  number of integers as input from the user. From these inputs, they need to calculate the sum of all even numbers, and that of all odd numbers, separately. These two values are then used to calculate their average. The function prints both the sums, as well as the average on separate lines. In the quest, when these values are written on the rock, the final obstacle will be removed, and the group may successfully come out of the Labyrinth.

#### Constraints

- $n \in \mathbb{N}$
- $n \geq 2$

#### Interaction

The input comprises of an integer,  $n$  representing the number of integers written on the rock, followed by a sequence of numbers separated by a space on the second line.

The output comprises of three values: the first line contains the value for even sum, the second line is the odd sum, and the last line is the average of the two sums.

**Sample**

| Input                          | Output |
|--------------------------------|--------|
| 4                              | 44     |
| 51 87 93 44                    | 231    |
|                                | 137.5  |
| 10                             | 256    |
| 11 21 31 42 52 62 73 83 93 100 | 312    |
|                                | 284.0  |

In the first case, there is only one even number in the sequence, so even sum = 44. The other three are odd numbers, so odd sum = 231. The final answer is the average of these two sums, which is also output on the screen = 137.5

In the second case, there are 4 even numbers, and their sum = 256. There are 6 odd numbers, and the odd sum = 312. The final output is their average value = 284.0

**Proposed Solution**

```
1      #Function Definition
2      def solve_labyrinth(N):
3          for 3m in range(n):
4              num = int(input())
5
6              if (num1 % 2 == 0):
7                  sum_even += num1
8
9          average = sum_even + (sum_odd/2)
10         print(sum_even)
11         print(avg)
12
13
```

**Dry Run**

Using any one of the inputs provided above, dry run your pseudocode in the space below.

**Error Identification**

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

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**Correct Solution**

Rewrite the lines of code you mentioned above with their errors corrected.

**Rough Work**